Hemanth Chowdary Kavula

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with **1+ years of experience** developing, maintaining, and optimizing **ETL/ELT pipelines** using **Python, SQL, PySpark, and AWS**. Hands-on experience with **Apache Spark, AWS Glue, Amazon EMR, Amazon S3, Snowflake, and dbt** for batch processing, data transformation, and cloud-based analytics. Skilled in **SQL development, dimensional data modeling, automated data quality validation, and pipeline troubleshooting**, with exposure to near-real-time data ingestion using **Apache Kafka and AWS Kinesis**. Experienced in preparing reliable financial datasets for **reporting, risk analytics, and downstream AI/ML applications**.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, Java, JavaScript
Cloud Platforms: AWS (S3, Glue, EMR, Redshift, Lambda, Kinesis)
Big Data & Streaming: Apache Spark, PySpark, Spark SQL, Apache Kafka, Delta Lake, Databricks
Databases & Data Warehousing: Snowflake, PostgreSQL, MySQL, SQL Server, Oracle
Data Integration & Orchestration: dbt, Apache Airflow, ETL/ELT Pipelines, Data Ingestion
Data Modeling & Quality: Dimensional Modeling, Star Schema, SCD Type 2, Great Expectations, Data Validation, Data Reconciliation
DevOps & Analytics: Git, Jenkins, Terraform, Docker, Power BI, Tableau
Machine Learning & Data Analysis: Pandas, Scikit-learn, Logistic Regression, CLV Modeling, BG/NBD, Gamma-Gamma

PROFESSIONAL EXPERIENCE

Fidelity Investments | Jersey City, NJ — Jan 2026 – Present

Data Engineer

- Developed and maintained **ETL/ELT pipelines using Python, SQL, PySpark, AWS Glue, and Amazon EMR** to process financial and trading datasets for downstream analytics and reporting.
- Engineered **PySpark transformations** to cleanse, filter, join, and aggregate raw data stored in **Amazon S3**, producing structured datasets for analytical workloads.
- Supported **near-real-time data ingestion using Apache Kafka and AWS Kinesis**, enabling trading event data delivery to downstream processing and monitoring systems.
- Developed **Snowflake SQL transformations and dbt models** to convert source data into analytics-ready tables supporting financial reporting.
- Automated **data quality validation using Python, SQL, and Great Expectations** to detect duplicate records, missing values, and schema inconsistencies before downstream data loading.
- Prepared and validated **Delta Lake datasets using PySpark**, supporting financial analytics and downstream AI/ML applications, including risk analysis and anomaly detection.
- Monitored **ETL pipeline executions**, investigated processing failures through log analysis and validation checks, and supported code deployments using **Git and Jenkins**.
- Optimized **SQL queries and PySpark transformations** through join optimization, predicate filtering, and workflow improvements to enhance pipeline execution efficiency and maintain reliable financial data delivery.

Mastercard | San Francisco, CA — May 2025 – Aug 2025

Data Engineer Intern

- Developed **Python and SQL scripts** under engineering team guidance to extract, clean, and transform structured datasets for internal analytics and reporting.
- Assisted with **ETL pipeline development, testing, and data loading**, validating source-to-target transformations across relational databases.
- Developed **SQL queries using JOINs, subqueries, aggregations, and filtering** to support data extraction, reporting, and transformation validation.
- Performed **data quality checks and SQL-based reconciliation** to identify missing values, duplicate records, and source-to-target discrepancies, documenting findings to support data accuracy.
- Supported **AWS S3 data processing and ETL troubleshooting**, collaborating with engineers to investigate data loading issues and document pipeline workflows.

PROJECTS

Customer Lifetime Value (CLV) & Churn Prediction | GitHub

- Designed a **MySQL star schema data warehouse** with transaction fact and customer, product, and date dimension tables; developed advanced SQL transformations using **CTEs and window functions** for RFM segmentation and feature engineering.
- Developed **Python-based CLV and churn prediction models** using Pandas, BG/NBD, Gamma-Gamma, and Logistic Regression; generated 6- and 12-month CLV estimates and evaluated churn predictions using ROC-AUC, precision, and recall to support customer retention analysis.

Technologies: Python, MySQL, SQL, Pandas, Scikit-learn, BG/NBD, Gamma-Gamma

Pizza Sales Analysis | GitHub

- Developed **14 SQL queries** across four normalized MySQL tables using JOINs, CTEs, subqueries, aggregations, and window functions to analyze sales transactions and revenue performance.
- Applied **SUM OVER and DENSE_RANK** to calculate cumulative revenue, identify top-performing pizzas by category, and analyze peak ordering hours and sales trends.

Technologies: MySQL, SQL, CTEs, Window Functions, Relational Data Modeling

Personal Portfolio Website | GitHub | Live Site

- Created and customized an **AI-assisted responsive portfolio website** using React.js, Vite, and Tailwind CSS to showcase Data Engineering experience, technical projects, and certifications.
- Integrated **GitHub project links, a downloadable resume, and Netlify-based deployment** to provide recruiters with access to professional credentials and project work.

Technologies: React.js, JavaScript, Vite, Tailwind CSS, GitHub, Netlify

EDUCATION

Master of Science in Computer Science Jan 2024 – Dec 2025
Rowan University | Glassboro, NJ

PUBLICATIONS

Age and Gender Prediction using Gated Residual Attention Network | IEEE (2023) | View Publication

CERTIFICATIONS

SQL (Advanced) Skill Certification — HackerRank | May 2025 | Verify
AWS Academy Cloud Foundations — AWS Academy | Feb 2022 | Verify
Foundations: Data, Data, Everywhere — Google / Coursera | Nov 2025 | Verify